

CMSI 630 – Artificial Intelligence

Instructor: Dr. K. Narayanaswamy

Assignment 1

AI-based Supervised Machine Learning Classification Model Selection

Due: 10/09/2026 @ 11:59:59pm

Assignment objective:

You are to write a program that will use an ML classification problem description and a provided training dataset to select the appropriate machine learning model to use for that problem. **We will use the IRIS dataset from our first class. It is small, already well-organized and documented.** The goal is to see which of many models might yield the best results for the IRIS dataset.

Additionally, your program should generate the code for the selected model, train it on the provided training dataset, and carry out validation of the selected model using the selected model, and present the performance of the model in terms of accuracy, precision, recall, etc., and that the selected model performs as well as or better than other classification models.

Assignment Requirements:

Make sure any code generated is Python. In the class examples, for demonstration purposes, we used Scikit Learn as the Machine Learning library for Python and Jupyter environment. It is best for the class to standardize on these libraries/environments for assignments to ensure that the assignments can be tested and graded without any additional installation problems. If you see technical issues that would make that hard for you, please let me know.

You may use any GenAI model as the foundation to analyze the problem and write the code needed. I have selected a small dataset to ease the problem. You can start off with the basic, free Anthropic code generation package to see if it satisfies the requirement. If experiences requires us to change paths, we can.

Your solution needs to explain the reasoning as to why the specific ML model was selected.

Suggested Assignment implementation strategy:

Using the GenAI model that you have elected to use to:

1. Select the best ML model for the problem description and the provided training dataset;
2. Generate the Python code needed to implement the selected model
3. Train the model on the provided database
4. Run the proper cross-validation methods using the training set and demonstrate its performance using methods like the confusion matrix, precision, recall and other functions available in the ML library that you have elected to use.

5. Provide documentation of the reasoning by which the specific model was selected from the problem description and training dataset

Other Guidelines:

1. Submit your assignment via Brightspace by the due date. No LATE SUBMISSIONS – you will get 0. Submit whatever you have by the deadline.
2. Make sure you document whatever GenAI APIs you are using.
3. Be prepared to demonstrate your program in class
4. Please test your program thoroughly on the systems mentioned above
5. You must write the prompts yourself. Borrowing someone else's prompts is cheating.

Grading Policy:

Total Points Available: 100

Your program will be graded on the following key code deliverables. Submit everything related to your code, prompts, outputs from the AI etc. to Brightspace.

1. Documentation of the overall design of your solution, platforms used and their roles, including how to run your code **(10 points)**
2. Selection of the best ML model for the problem from description/training data: **20 points**
3. Jupyter module that includes the implementation of the selected ML model: **20 points**
4. Jupyter module running the cross-validation procedures needed to demonstrate performance: **20 points**
5. Documentation of the reasoning for ML model selection and its performance relative to the other possible models: **15 points**
6. 5-minute In-class demo and explanation: **15 points**

You can submit a single Jupyter that includes ALL the above. Please label the items to make them easier to find.